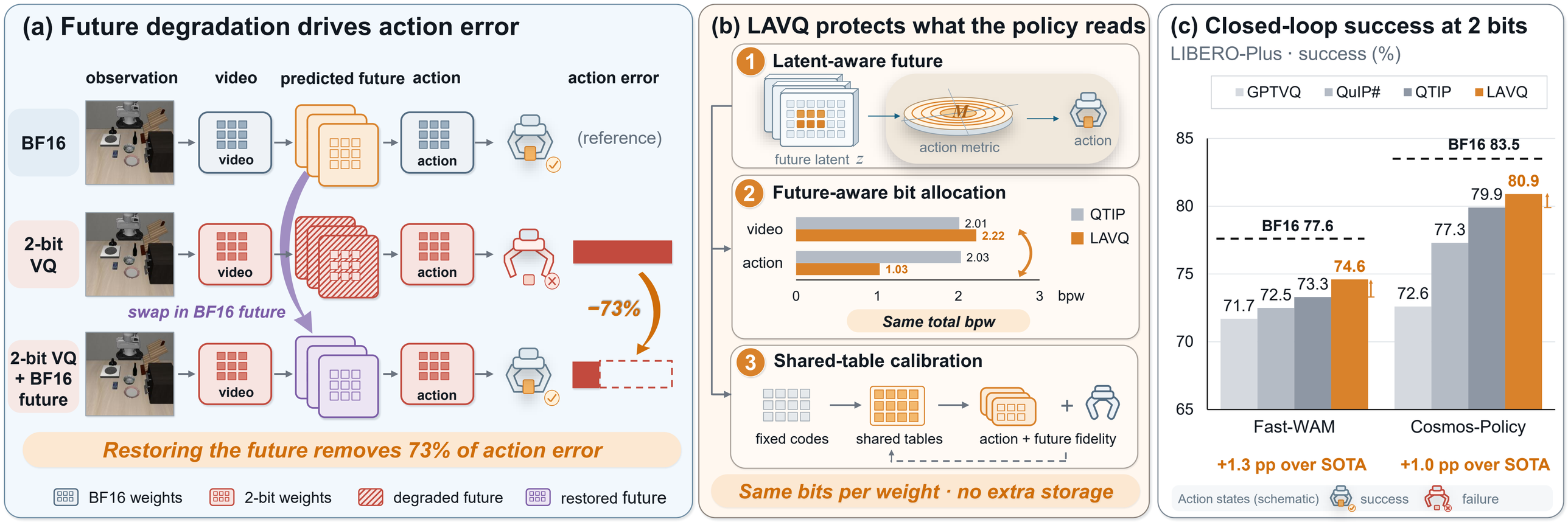

(a) Future degradation drives action error
observation
video
predicted future
action
action error
BF16
video
action
(reference)
2-bit VQ
video
action
swap in BF16 future
−73%
2-bit VQ + BF16 future
video
action
Restoring the future removes 73% of action error
BF16 weights
2-bit weights
degraded future
restored future
(b) LAVQ protects what the policy reads
1 Latent-aware future
future latent z
M
action metric
action
2 Future-aware bit allocation
video
action
2.01
2.22
2.03
1.03
QTIP
LAVQ
0
1
2
3
bpw
Same total bpw
3 Shared-table calibration
fixed codes
shared tables
action + future fidelity
Same bits per weight · no extra storage
(c) Closed-loop success at 2 bits
LIBERO-Plus · success (%)
GPTVQ
QuIP#
QTIP
LAVQ
85
80
75
70
65
BF16 83.5
BF16 77.6
71.7
72.5
73.3
74.6
72.6
77.3
79.9
80.9
Fast-WAM
Cosmos-Policy
+1.3 pp over SOTA
+1.0 pp over SOTA
Action states (schematic)
success
failure